DESIGN AND ANALYSIS OF ALGORITHMS (20MCA203)

MODULE-I

3 MARKS

1. Algorithm & it's properties
2. Differentiate Time & Space complexity
3. Divide and Conquer control abstraction
4. Strassen's Matrix Multiplication
5. Solve the following recurrence equation using iteration method.
6. T(n)=2T (n/2) + n

6 MARKS

1. Describe different Asymptotic notations used for expressing time complexity of algorithms.
2. Explain the concept of Merge Sort algorithm with an example.
3. Write the algorithm for Quick Sort and sort the given set of elements using it

MODULE-II

3 MARKS

1. State and illustrate the Principle of Optimal Substructure
2. Explain the Greedy Control abstraction.
3. What are the steps to follow in order to solve a problem using Dynamic programming?
4. Write a note on MST
5. Discuss about 'All Pair shortest path problem' in detail.
6. Explain the pseudo code of Bellman ford algorithm for finding the single shortest path.

6 MARKS

1. Explain the different steps used to find the minimum cost spanning tree for the given graph using Prim's algorithm.
2. Solve TSP for the given graph

3. Job sequencing problem
4. Explain Knapsack problem. Find out the optimal solution n=5, (P1,P2… ..,P5)=(6,14,6,9,5), (w1,w2,… ..w5)=(5,7,2,3,4) and m=15 using Fractional knapsack.
5. Apply Floyd-Warshall algorithm to find all-pairs shortest path problem.

MODULE-III

3 MARKS

1. How does Backtracking differ from Branch and Bound?
2. Explain the concept of State space Tree in B&B.
3. Apply backtracking technique for solve the following instance of Subset sum

   problem w={3,4,5,6) and d=9.
4. State-space tree for 4-Queens problem
5. Discuss about subset sum problem.
6. Write a comparison-based sorting algorithm used by lower bound theory technique.
7. Bring out the notion of Decision Trees.

6 MARKS

1. Discuss the sum of subsets algorithm with the following example A = {1, 2, 5, 6, 8}, Find the all-possible combinations whose sum is equal to the given value M = 9.
2. Explain the N-Queen's problem and its solution using Backtracking.
3. Discuss about N2 -1 puzzle problem

MODULE-IV

3 MARKS

1. What do you mean by the term Polynomial time reduction?
2. Explain P & NP-Class
3. Explain Clique problem with an example.
4. Prove that Clique is NP-Complete
5. Differentiate NP Hard and NP complete problem.

6. Explain the Bipartite matching problem with an example.

6 MARKS

1. Compare Clique and Vertex cover problems.
2. Compute the maximum flow in the given network using Ford- Fulkerson algorithm.
3. Discuss Complexity classes. Prove that Vector cover problem is NP Complete.

MODULE-V

3 MARKS

1. Describe approximation ratio in approximation algorithm.
2. Write note on approximation algorithm.
3. What is randomized algorithm?
4. What are the two different types of Randomized algorithms?

6 MARKS

1. Explain the 2-Approximation algorithm for vertex cover and justify its approximation ratio.
2. Vertex Cover Approximation using Linear Programming and LP Rounding Algorithm.
3. What is Randomised Quick sort? Explain its working and why is it used